

Chemistry, 2018 EC, entrance exam, round 1

Grade 12 · Chemistry · 2018 E.C.

Time allowed: 120 minutes

Questions: 17

Total marks: 17

Name _____ Grade _____ ID No. _____ Date _____

1. The masses of different substances, liberated or deposited by the same amount of electricity, are proportional to their equivalent masses. This statement is _____. (1)
- A. Faraday's first law of electrolysis
B. the law of definite proportion
C. Faraday's second law of electrolysis
D. the law of multiple proportion
2. Which of the following reaction represents the chemical equation for the reaction that takes place between aluminum metal and solution of hydrochloric acid? (1)
- A. $2\text{AlCl}_3(\text{aq}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{Al}(\text{s}) + 6\text{HCl}(\text{aq})$
B. $2\text{Al}(\text{s}) + 6\text{HCl}(\text{aq}) \longrightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2(\text{g})$
C. $2\text{Al} + 6\text{HCl}(\text{aq}) \longrightarrow 2\text{AlCl}_3(\text{aq})$
D. $\text{Al} + \text{HCl} \longrightarrow \text{AlCl}_3$
3. A cell containing AlCl_3 as an electrolyte, is placed in series with another cell containing AgNO_3 as an electrolyte. What mass of Ag is deposited, when 18 g of Al is deposited at the cathode electrode? ($M_{\text{Ag}} = 108 \text{ g/mol}$ and $M_{\text{Al}} = 27 \text{ g/mol}$) (1)
- A. 36 g
B. 24 g
C. 216 g
D. 72 g
4. Nitric acid is a colorless fuming liquid with a boiling point of 83°C . However, it turns yellow as it stays for long time. Which of the following accounts for the observed property of nitric acid? (1)
- A. NO_2
B. NO
C. NO_2^-
D. NO_3^-
5. Which of the following tells the exact number of atoms of different elements present in a molecule? (1)
- A. Molecular mass
B. Empirical formula
C. Formula mass
D. Molecular formula
6. What is the concentration of hydrogen ion in a solution of 0.1 M nicotinic acid, $\text{HC}_6\text{H}_4\text{NO}_2$, at 25°C ? ($K_a = 1.4 \times 10^{-5}$) (1)
- A. $1.2 \times 10^{-5} \text{ M}$
B. $1.2 \times 10^{-3} \text{ M}$
C. $1.4 \times 10^{-5} \text{ M}$
D. $1.4 \times 10^{-6} \text{ M}$
7. Chlorine has two naturally occurring isotopes. In a sample of chlorine, 25% of the atoms are Cl-37 and 75% of the atoms are Cl-35. What is the average atomic mass of chlorine? (1)
- A. 26.25
B. 25.75
C. 35.5
D. 36.0

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8. A student performs an experiment to distinguish different oxides by doing the following experiment. He/she put 100 mL KI solution in to four different test tubes. He/she acidified each of these solutions with 10 drops of dilute H_2SO_4 solution. He/she added a spatula full of CaO in the first, Al_2O_3 in the second, Na_2O_2 in the third and P_4O_{10} in the fourth test tubes. Finally, he/she added 10 mL starch solution to each of the four test tubes. In which test tube will he/she observe a color change? In the test tube that contains _____.
- A. Test tube 2
B. Test tube 3
C. Test tube 4
D. Test tube 1
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9. Combustion of acetylene at about 3000°C with oxygen produces an intense hot flame. In which group of hydrocarbons is this compound classified?
- A. Alkanes
B. Aromatics
C. Alkenes
D. Alkynes
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10. Phosphorus pentafluoride, PF_5 is one example of exception to the octet rule. Why does PF_5 is considered to be exceptional to the octate rule? This is because the _____.
- A. central atom has more than eight valence electrons.
B. atoms attached to P have more than eight valence electrons.
C. central atom has less than eight valence electrons.
D. central atom has eight valence electrons.
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11. You are supposed to test the conductivity of the substances using a conductivity apparatus as shown in the figure below: Which of the following aqueous solutions does NOT cause the bulb to glow when the switch turns on?
- [Figure: A schematic diagram of a conductivity apparatus. It shows a rectangular container filled with a light blue liquid labeled "Aqueous solution". Two vertical black electrodes are partially submerged in the solution. Above the container, a yellow electric bulb is connected to the system, with a label "Bulb" pointing to it. Below the container, there is a green battery icon labeled "DC source or Battery".]*
- A. Sodium hydroxide solution
B. Copper sulphate solution
C. Table salt solution
D. Sugar solution
-
12. Copper metal reacts with sulfuric acid to produce copper sulphate, sulfur dioxide and water. Which of the following is the coefficient of H_2SO_4 after balancing this chemical reaction?
- A. 6
B. 1
C. 2
D. 3
-
13. What is the amount of energy emitted when an electron moves from the $n = 4$ to $n = 2$ energy level? ($R_H = 2.18 \times 10^{-18} \text{ J}$)
- A. $-6.54 \times 10^{-18} \text{ J}$
B. $-4.09 \times 10^{-19} \text{ J}$
C. $4.09 \times 10^{-19} \text{ J}$
D. $6.54 \times 10^{-18} \text{ J}$
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14. Consider the following diagram used for the electrolytic refining of copper What is the name of species indicated by letter 'C'?
- [Figure: A diagram of an electrolytic refining cell with a switch connected to two electrodes suspended in a tank containing an electrolyte solution labeled A. The left electrode is a yellow vertical bar labeled Cathode. The right electrode is a brown vertical bar labeled Anode. At the bottom of the tank beneath the anode, there is a dark deposit labeled C (Settled Impurities) with a letter B pointing to a small rectangular block resting on it near the bottom.]*
- A. Copper sulphate solution
B. Strip of pure copper
C. Cathode mud
D. Anode mud

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15. Consider the reaction given below: $AB_2C + C \longrightarrow AB_2 + C_2$ Which of the following successful collisions are effective in the formation of the products? (1)
- A. The collision that results in the A and C atoms are directly facing each other.
 - B. The collision that results in the C atoms are directly facing each other.
 - C. The collision that results in the B and C atoms are directly facing each other.
 - D. The collision that results in the A and B atoms are directly facing each other
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16. At 0 °C and 1.0 atm, the solubility of gas in water is 48.4 mole per liter. What is the molar solubility of in a saturated water solution at a partial pressure of 0.20 atm? (1)
- A. 1.20×10^{-3} M
 - B. 9.68 M
 - C. 4.10×10^{-3} M
 - D. 2.44 M
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17. Which one of the following does NOT explain the chemical properties of acidic oxides? Acidic oxides react with: (1)
- A. acids to form salt and water.
 - B. basic oxides to form salts.
 - C. water to form acids.
 - D. base to form salt and water.

— END OF EXAMINATION —